

DAY 15 — DOCKER (FINAL VERSION)

GOAL

You will convert your entire system into:

Portable AI system (run anywhere)

WHY THIS MATTERS

Right now: Runs only on your machine (manual setup)

After today: One command → full system runs

✅ STEP 1 — CREATE DOCKER FILE

📍 Location: **PROJECT ROOT**
(same level as app/, requirements.txt)

Dockerfile

```
FROM python:3.10-slim
```

```
# Set working folder
```

```
WORKDIR /app
```

```
# Copy all files
```

```
COPY . .
```

```
# Install dependencies
```

```
RUN pip install --no-cache-dir -r requirements.txt
```

```
# Expose API port
```

```
EXPOSE 8000
```

```
# Run FastAPI
```

```
CMD ["uvicorn", "app.api:app", "--host", "0.0.0.0", "--port", "8000"]
```

✅ STEP 2 — CREATE .dockerignore

📍 Location: **PROJECT ROOT**

 .dockerignore

__pycache__/

*.pyc

*.pyo

.env

logs/

data/db/

 STEP 3 — IMPORTANT FIX (LOCAL MODELS)

 **YOU ARE USING:**

Ollama (Qwen)

 Docker container **cannot access your host model directly**

 OPTION (REQUIRED FOR NOW)

Run container using **host network**

 STEP 4 — BUILD IMAGE

 Run in root:

docker build -t multimodal-ai .

 STEP 5 — RUN CONTAINER

 Linux / Mac

docker run --network=host multimodal-ai

 Windows

docker run -p 8000:8000 multimodal-ai

✅ STEP 6 — VERIFY SYSTEM

Open: <http://127.0.0.1:8000/docs>

Test:

POST /ask { "query": "average sales" }

✅ STEP 7 — DEBUG (IF ERROR)

If LLM not responding:

👉 Ensure: ollama serve

running on host

✅ FINAL FLOW

Docker Container

↓

FastAPI

↓

Agent System

↓

Ollama (host)

↓

Response

✅ YOUR FOLDER STRUCTURE NOW

mutimodel_llm/

|

├─ app/

├─ embeddings/

├─ retrieval/

└─ tools/

└─ ingestion/

└─ models/

└─ memory/

└─ evaluation/

|

└─ logs/

└─ data/

|

└─ Dockerfile  NEW

└─ .dockerignore  NEW

└─ requirements.txt

TEST FULL SYSTEM

Try queries:

>> what is ai

>> summarize document

>> average sales

>> compare trends

WHAT YOU BUILT TODAY

Portable AI backend system

SYSTEM LEVEL NOW

Before:

Local app

Now:

Containerized AI service

✅ COMMON MISTAKES (IMPORTANT)

- ❌ Forget to install requirements
- ❌ Ollama not running
- ❌ Wrong path in API
- ❌ Not exposing port

✅ DAY 15 RESULT

You now have:

- ✅ AI system
- ✅ API
- ✅ Docker (portable)

✅ NEXT

DAY 16 → Async + Parallel Execution

If ready:

👉 say **Day 16**